

PROJECT OUTPUTS & DIVISIONS

Smart Digital Closet and Outfit Recommendation System



**Submitted To:
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PROJECT DIVISIONS:

1. Urneeb Zahra

Module: User Authentication & File Handling

Responsibilities:

- Signup System
- Login System
- User Verification
- Register User Function
- File Handling for Users
- Save Clothes to File
- Load Clothes from File
- Extract User Function
- Data Management using `fstream`

Functions Covered:

- `signupMenu()`
- `loginMenu()`
- `registerUser()`
- `verifyLogin()`
- `userExists()`
- `saveClothesToFile()`
- `loadClothesFromFile()`
- `extractUser()`

Concepts Used:

- File Handling
- Arrays
- String Manipulation
- Validation
- Data Persistence

2. Muhammad Saad Asghar

Module: OOP Design & Clothing Management

Responsibilities:

- Designing Base Class and Derived Classes
- Inheritance & Polymorphism
- Add Clothing Features
- Edit Clothing Features
- Delete Clothing Features
- View Closet System
- Menu Design & User Interface

Classes Covered:

- ClothingItem
- Upper
- Lower
- Shoes

Functions Covered:

- `addUpper()`
- `addLower()`
- `addShoes()`
- `viewAllClothes()`
- `editClothing()`
- `deleteClothing()`
- `addClothingMenu()`
- `closetMenu()`

Concepts Used:

- OOP
- Inheritance
- Polymorphism
- Encapsulation
- Arrays of Objects
- Menu-driven Programming

3. Ahmed bin Kashif Malik

Module: Outfit Recommendation System & Matching Logic

Responsibilities:

- Outfit Recommendation Logic
- Color Matching System
- Shoe Validation Logic
- Season & Occasion Matching
- Recommendation Output Display
- Main Program Flow

Functions Covered:

- `colorsMatch()`
- `shoeValid()`
- `recommendOutfits()`
- `printLine()`
- `printDash()`
- `main()`

Concepts Used:

- Conditional Statements
- Nested Loops
- Matching Algorithms
- Recommendation Logic
- User Interaction
- Console Formatting

OUTPUT SCREENSHOTS:

```
=====
SMART DIGITAL CLOSET & OUTFIT RECOMMENDATION
=====
1. Login
2. Signup
0. Exit
-----
Choice: 1

=====
USER LOGIN
=====
Username : saad
Password : 1234

[SUCCESS] Welcome, saad!

=====
DIGITAL CLOSET MENU
User: saad
=====
```

```
1. Add Clothing
2. View My Closet
3. Edit a Clothing Item
4. Delete a Clothing Item
5. Get Outfit Recommendations
0. Logout
-----
Choice: 1

=====
ADD CLOTHING TO YOUR CLOSET
=====
1. Add Upper Wear (Shirt, Kameez, Jacket...)
2. Add Lower Wear (Jeans, Shalwar, Trousers...)
3. Add Shoes
0. Back
-----
Choice: 1
-----
ADD UPPER WEAR
-----
Item Name (e.g. Shirt, Kameez)      : Black Shirt
Color (e.g. Blue, White, Black)     : Black
Gender (Boys / Girls / Unisex)      : Boys
```

Choice: 1

ADD UPPER WEAR

Item Name (e.g. Shirt, Kameez) : Black Shirt
Color (e.g. Blue, White, Black) : Black
Gender (Boys / Girls / Unisex) : Boys
Fabric (Cotton/Wool/Polyester/Silk) : Cotton
Season (1=Summer 2=Winter 3=Both) : 1
Occasion (1=Wedding 2=Sports 3=Formal
4=Casual 5=Party 6=Univ) : 3

[SUCCESS] Upper wear added!

=====

DIGITAL CLOSET MENU

User: saad

=====

Choice: 3

=====

EDIT CLOTHING ITEM

=====

Select category to edit:

- 1. Upper Wear (3 items)
- 2. Lower Wear (2 items)
- 3. Shoes (1 items)
- 0. Back

Choice: 1

=== UPPER WEAR ===

[1] Shirt | hite | otton

[2] Black Shirt | Black | Cotton

[3] white shirt | white | cotton

Enter item number to edit (0 to cancel): 1

Enter item number to edit (0 to cancel): 1

Editing: Shirt

(Press Enter to keep current value)

New Name [current: Shirt]: White Shirt

New Color [current: hite]: White

New Gender [current: oys]: Boys

New Fabric [current: otton]: Cotton

Season (1=Summer 2=Winter 3=Both) [current: 1]: 1

Occasion (1=Wedding 2=Sports 3=Formal 4=Casual 5=Party 6=Univ) [current: 3]: 3

[SUCCESS] Item updated!

[UPPER WEAR]

Name : White Shirt

Color : White

Gender : Boys

Season : Summer

Occasion : Formal

Category : Upper

Fabric : Cotton

=====

DIGITAL CLOSET MENU

User: saad

=====

1. Add Clothing
 2. View My Closet
 3. Edit a Clothing Item
 4. Delete a Clothing Item
 5. Get Outfit Recommendations
 0. Logout
-

Choice: 5

=====

OUTFIT RECOMMENDATION SYSTEM

=====

Select Occasion:

1=Wedding 2=Sports 3=Formal 4=Casual 5=Party 6=University

Your choice: 3

Select Season: 1=Summer 2=Winter 3=Both

Your choice: 1

=====

Your choice: 3

Select Season: 1=Summer 2=Winter 3=Both

Your choice: 1

=====

RECOMMENDED OUTFITS

=====

>> OUTFIT #1 <<

UPPER : White Shirt (White, Cotton)
LOWER : Trousers (Black, Cotton)
SHOES : Oxford (Black, Formal)
SEASON : Summer
FOR : Formal Occasion

>> OUTFIT #2 <<

UPPER : White Shirt (White, Cotton)
LOWER : Jeans (Blue, Denim)
SHOES : Oxford (Black, Formal)
SEASON : Summer
FOR : Formal Occasion

>> OUTFIT #2 <<

UPPER : White Shirt (White, Cotton)
LOWER : Jeans (Blue, Denim)
SHOES : Oxford (Black, Formal)
SEASON : Summer
FOR : Formal Occasion

>> OUTFIT #3 <<

UPPER : Black Shirt (Black, Cotton)
LOWER : Trousers (Black, Cotton)
SHOES : Oxford (Black, Formal)
SEASON : Summer
FOR : Formal Occasion

>> OUTFIT #4 <<

UPPER : Black Shirt (Black, Cotton)
LOWER : Jeans (Blue, Denim)
SHOES : Oxford (Black, Formal)
SEASON : Summer
FOR : Formal Occasion


```
>> OUTFIT #5 <<
```

```
-----  
UPPER   : white shirt (white, cotton)  
LOWER   : Trousers (Black, Cotton)  
SHOES   : Oxford (Black, Formal)  
SEASON  : Summer  
FOR      : Formal Occasion
```

```
>> OUTFIT #6 <<
```

```
-----  
UPPER   : white shirt (white, cotton)  
LOWER   : Jeans (Blue, Denim)  
SHOES   : Oxford (Black, Formal)  
SEASON  : Summer  
FOR      : Formal Occasion  
=====
```

```
Total Outfits Found: 6  
=====
```

```
=====
```

```
DIGITAL CLOSET MENU  
User: saad  
=====
```

- ```
1. Add Clothing
2. View My Closet
3. Edit a Clothing Item
4. Delete a Clothing Item
5. Get Outfit Recommendations
0. Logout

```

```
Choice: 4
=====
```

```
DELETE CLOTHING ITEM
=====
```

- ```
Select category to delete from:  
1. Upper Wear (3 items)  
2. Lower Wear (2 items)  
3. Shoes (1 items)  
0. Back  
-----
```

Choice: 1

=== UPPER WEAR ===

[1] White Shirt | White | Cotton

[2] Black Shirt | Black | Cotton

[3] white shirt | white | cotton

Enter item number to delete (0 to cancel): 3

You are about to delete: white shirt

Are you sure? (1=Yes 0=No): 1

[SUCCESS] Item deleted successfully!

=====

DIGITAL CLOSET MENU

User: saad

=====